

Shubhomoy Sarkar

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PROFESSIONAL SUMMARY

Computer Science Engineering student at IIIT Kottayam building at the intersection of full-stack architecture, generative AI, and multi-agent systems. Experienced in designing production-ready applications, low-latency reasoning pipelines, and scalable backend workflows that take intelligent systems out of prototypes and into high-throughput live environments. Passionate about systems efficiency, clean code design, and advanced algorithmic optimization.

EDUCATION

Indian Institute of Information Technology Kottayam B.Tech in Computer Science and Engineering; CGPA: 8.24/10	Kottayam, India 2024 – 2028
The Modern Academy, Kolkata Class 12 ISC (PCMB): 88% Class 10 ICSE: 94%	Kolkata, India 2024 2022

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, SQL, Java, MATLAB
- **Generative AI & LLMs:** Google Gemini API, Google Agent Development Kit (ADK), Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Tool Calling, Function Calling, Structured Outputs, Context Engineering, Serper API
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Sentence Transformers, XGBoost, CNNs, ANNs, NumPy, Pandas, SciPy, Node2Vec, Learning-to-Rank, Model Optimization
- **Software Engineering:** React, TypeScript, Express.js, Flask, REST APIs, Node.js, PostgreSQL, Git, Linux, PyQt5, CMake
- **Computer Vision & Applied AI:** YOLO, OpenCV, Image Processing, OpenGL, Camera Geometry, Rendering Pipelines, Feature Extraction
- **Core Computer Science:** Data Structures & Algorithms (Advanced Complexity Optimization), Operating Systems, DBMS (Concurrency, ACID Properties, Indexing), Computer Networks, Directed Acyclic Graphs (DAG), Kahn's Algorithm.

EXPERIENCE

Generative AI Engineering Intern Decodelabs	May 2026 – June 2026 Bengaluru, India / Remote
- Engineered production-grade Generative AI systems, deploying context-aware memory chatbots and automated multi-language code review pipelines with high structural reliability. - Architected intelligent memory buffers and secure multi-modal workflows using Python and the Google Gemini API, ensuring robust state management and token optimization across user sessions. - Designed modular software architectures to seamlessly integrate LLMs into live production environments, prioritizing ultra-low latency, input sanitization, and system security. - Collaborated with cross-functional engineering teams to debug SDK integrations, context handling mechanisms, and API communication layers for real-time applications.	

PROJECTS

Bodd AI Workspace – GenAI & Graph Theory <i>React 18, Express.js, Gemini SDK, React Flow, Kahn's Algo</i>	
- Engineered a next-generation AI workspace that replaces traditional linear chat interfaces with a Directed Acyclic Graph (DAG) topology for advanced, complex reasoning workflows. - Integrated Kahn's Algorithm to manage state dependencies, enforce execution order, and optimize the underlying asynchronous AI node execution network. - Utilized React Flow and Express.js to build a responsive, highly interactive full-stack interface for visualizing complex AI thought processes and execution paths in real-time.	
SentinelAI – IoT & Edge AI	<i>ESP32-CAM, YOLO11n, ByteTrack, Qwen2.5-VL, Llama 3</i>

- Developed an end-to-end edge surveillance architecture that upgrades a standard ESP32-CAM hardware module into a proactive, intelligent AI security assistant.
- Integrated YOLO11n and ByteTrack algorithms for high-precision, real-time object detection, classification, and trajectory tracking directly on edge-processed video streams.
- Leveraged the advanced multi-modal reasoning capabilities of Qwen2.5-VL and Llama 3 to analyze visual anomalies, interpret scene contexts, and trigger automated, context-aware alerts.

GreenTravel API – ML & Process Mining

Python, Scikit-Learn, Celonis, Pandas, PostgreSQL

- Analyzed large-scale historical corporate travel datasets to pinpoint structural process bottlenecks, inefficiencies, and enterprise transit optimization opportunities.
- Developed predictive machine learning models using Scikit-Learn to enforce strategic transport modal shifts and quantitatively reduce organizational carbon footprints.
- Built robust data engineering pipelines with Pandas and PostgreSQL, integrating with Celonis for advanced process mining, event-log analysis, and data visualization.

Decodelabs Suite – Production AI Suite

Python, Gemini API, Modular Architecture, Security Pipelines

- Built a comprehensive suite of production-grade generative AI applications spanning multi-turn conversational agents and automated multi-language code review engines.
- Designed strictly typed security pipelines to sanitize inputs, manage dynamic context windows, and safeguard multi-modal LLM interactions in live production deployments.

ACHIEVEMENTS & CERTIFICATIONS

- **Cloud with Anthropic API Certification:** Validated expertise in building cloud-integrated AI workflows using Claude models and advanced Anthropic API architectures (Verified).
- **AWS Cloud Practitioner Essentials:** Gained comprehensive foundational knowledge of AWS cloud services, infrastructure security, pricing models, and cloud deployment strategies (Verified).
- **Smart India Hackathon 2024:** Successfully qualified the rigorous internal selection round for India's premier nationwide innovation and software hackathon initiative.
- **ML Workshop – Techfest, IIT Bombay (2025):** Completed intensive technical training and hands-on machine learning applications during IIT Bombay's annual tech festival.
- **Summer Analytics – IIT Guwahati (2025):** Participated in a highly competitive data science and analytics program covering statistical modeling, data cleaning, and analytical tools.

LEADERSHIP & POSITIONS OF RESPONSIBILITY

- **PR & Event Management Sublead (Trendles):** Directed institutional public relations, managed large-scale event logistics, and spearheaded digital promotional campaigns and sponsor communications (Oct 2025 – Present).
- **E-Cell Core Member:** Collaborated on campus-wide entrepreneurial initiatives, fostering student startup growth, organizing structured pitch events, and supporting institutional innovation frameworks.